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Dr Qiongxu Li and the assessment committee
Department of Electronic Systems, Technical Faculty of IT and Design
Aalborg University Copenhagen, Denmark

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Re: PhD in Secure and Privacy-Preserving Collaborative Language Models (ref. 2026/987)

Dear Dr Li,

I once used a large language model's own explanations to teach cheaper models to label the way it does, for about \$12.50 in API calls. At Artemis AI, the UK start-up where I am co-founder and CTO, we needed training data for six text classifiers. I designed a chain in which Claude labelled about 6,000 public posts and, because we asked it to, explained each label in its response; we mined the decision rules out of those written explanations into a prompt for DeepSeek, which labelled 12,000 more at a fraction of the cost, and our small encoder models were trained on the result. It worked, and it was done through paid access for our own classifiers. But step for step it is the pipeline of a model-extraction attack: query, collect the outputs and the explanations, imitate. The explanations were the part that made it cheap. Your project asks how several parties can build on language models together while keeping their data confidential and the models intact. That is the question I want to work on for the next three years.

Why this PhD, and why your group. Because of that experience, the posting's line on defences against model extraction and chain-of-thought attacks led me straight to your preprint with Luo, Ren, Yu, Li and Bjerva, *Capable yet Parsimonious* (arXiv:2609.26637). You induce frontier models to externalise their hidden reasoning by registering a simple custom tool through a standard API feature, and because such traces might be post-hoc rationalisation, you first test them against native chain-of-thought on open models and find that they match its performance. I never tested the faithfulness of Claude's explanations before mining rules from them. Your mitigations appendix suggests restricting forced tool selection or inspecting tool arguments, but notes that legitimate planning tools produce similar content, so a provider also has to tell legitimate tool use apart from systematic extraction. That is a detection problem I know from the collection side, as I explain below. Your EMNLP 2026 paper with Junjie Mu on routing hijacking in federated RAG (arXiv:2605.28112) shaped the collaborative part of my plan. A malicious client forges its profile to attract target queries, and the defence has to work without trusting what participants say about themselves. I see the same structure in the problem my own work points to: organisations that each check an LLM annotator against their own small hand-labelled samples would gain a great deal by pooling that evidence, but they cannot share the samples and cannot assume every party is honest. The enclosed project description proposes to study private joint validation of shared LLM annotators, what the shared statistics leak, and how to detect extraction, including chain-of-thought extraction, spread across many accounts. The posting's title echoes your DFF project WHISPERS on privacy-preserving collaborative language models via topological federated learning; in my plan that is RQ2, fine-tuning over graphs of organisations where topology decides what colluding nodes can see, and I would weight it more heavily if the group needs that.

Python and distributed systems at scale. Python is my working language in production. At M+C Saatchi Fluency, where I have been a part-time Data Engineer since December 2024, I designed and operate a fleet of 64 Scrapy forum crawlers for a global automotive brand behind a CI gate and alerting, and I operate and repair a weekly AWS pipeline (Lambda, Fargate, Postgres) whose master table holds 2.2 million rows by 184 columns for a global e-commerce client, including memory fixes and an 89-check data-quality sweep before each release. At Fix.com, a contract role since May 2023, I have designed and run its crawler platform for parts catalogues on my own since June 2026 (about 10 million rows across 82 datasets by October). At Dubizzle Labs, Lahore (February 2022 to August 2024, Associate Software Engineer to Senior Data Engineer), I led a three-engineer team running more than 500 scrapers across 15 markets on Kubernetes/EKS. None of this is distributed training, but it is what collaborative learning runs on: many independent workers, unreliable inputs, and failures caught before they reach the data.

LLM pipelines and their validation. At Artemis AI, where I lead the technology in a team of eight ahead of pilots in December 2026, our classifiers learn from two LLM labelling tiers, first checked against about 500 posts and comments the team labelled by hand. When I compared the tiers, they agreed on 82.9% of sentiment and 74.4% of emotion labels but on no toxicity labels at all, because they used incompatible taxonomies, while the toxicity model

reported high confidence. A model-quality audit I ran in September found that our reported accuracies measured agreement with the labelling model rather than with people, so we are rebuilding evaluation around a human-labelled golden set before launch. At Fluency, on a study of twelve months of social discourse, every LLM tag records a hash of the prompt version that produced it, and three pilot runs on the same 500 rows were compared before the full run; I also contribute to LLM-assisted coding of about 170,000 multilingual posts and reviews for a global infant-nutrition brand.

Securing agents that read the open web. The posting's strand on secure agentic AI describes systems I already run. For the crawler fleet I designed a self-repairing agent: when a source changes shape it diagnoses the break, escalates between Claude models by cause, regenerates the spider and tests it against live pages. It reads untrusted third-party text and writes code that runs in production, so I treat it as an attack surface: page content enters the prompt fenced as untrusted data, secrets and cookies are refused and caught by a scanner, generated code is checked against forbidden patterns, every merge needs a single-use approval code tied to the test sample and expiring in ten minutes, and the build service colleagues' agents call sits behind OAuth with PKCE. At Artemis AI I am now leading the build of Art-E, an advisor that answers each client's questions from that client's own data, for pilots in December. The approaches I am applying are engineering ones: tenants isolated at the database, retrieval scoped to one client's records, prompt injection red-teamed before release, and an output check that rejects any answer naming a number absent from that client's data. They are rules, not guarantees. What this PhD would give that work is the part I cannot yet supply: proofs of what shared statistics and model updates reveal, ways for clients to pool evidence without pooling data, and detection of extraction through an agent's own tools.

Research, and what I know about extraction. My research so far is about measurement validity: whether a machine label means what it claims. I took my MSc in Data Analytics at the University of Portsmouth as a Chevening Scholar (2024/25, Distinction). My dissertation analysed about 279,000 public posts on UK economic policy from four platforms under university ethics approval, and I hand-labelled 200 posts myself to check the lexicon labels (kappa about 0.78 against VADER). With my supervisor Dr Ella Haig I developed it into a methods manuscript, *Is the Platform Part of the Measurement?*, which brings psychometric tools to machine-labelled text and shows in a 500-replication simulation that pooling platform scores at face value gives the wrong sign of aggregate sentiment on roughly a quarter of days. It is complete and not yet submitted; the code is public (github.com/asadfix/platform-measurement, DOI 10.5281/zenodo.23105129). This is why your ACL 2026 paper with Luo, Chen and Bjerva on cue-controlled PII memorisation struck me: it shows that much reported leakage is explained by cues in the prompt, which is a validity argument about a privacy metric. The other thing I bring is practical knowledge of query-based extraction. Since 2018 I have built automated web-data collection professionally (300+ completed Fiverr orders, then production crawlers for sites behind commercial bot-management systems). That work taught me how defenders recognise automated collection: request rate and timing, the structure and repetition of requests, client and browser fingerprints, and challenge pages. Extracting a model or its reasoning through an API is the same problem in a new place, and I want to work on the defender's side of it.

What I do not bring yet. I have no background in cryptography, secure multiparty computation or homomorphic encryption, and I have not implemented federated learning or differential privacy. The models I fine-tune are encoder classifiers; I use Claude and DeepSeek through APIs and have not fine-tuned a generative model. I have no peer-reviewed publications yet. I would plan the first year around these gaps: the doctoral school's courses, reproducing a standard federated fine-tuning baseline with differential privacy before extending it, and working through secure aggregation and secret sharing with your group, starting from its own work on privacy-preserving distributed processing and on attacks against secure aggregation. My BSc in Computer Science (FAST-NUCES, August 2017 to January 2022) included probability and statistics, linear algebra and numerical methods, and the mathematics I have used most since is statistical: the bootstrap and McNemar tests of my dissertation and the psychometric protocol and 500-replication simulation of my manuscript. I would add optimisation and information theory in the first year, since your group's privacy analysis rests on them.

I can start on 1 November 2026 or soon after and would relocate to Copenhagen. Enclosed are my CV, my BSc and MSc diplomas with grades, and a two-page project description. Dr Ella Haig and Dr Fahad Ahmad (both University of Portsmouth) have agreed to act as referees. Thank you for considering my application.

Yours sincerely,

Asad Ikram